SALES PERFORMANCE DASHBOARD

Excel-Based Data Analytics Project

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Field: Data Science
Tools Used: Microsoft Excel
Reporting Period: 1 January 2025 – 31 December 2025

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1. Executive Summary

This project involved developing an interactive Sales Performance Dashboard using Microsoft Excel. The purpose was to transform structured sales records into a visual summary of business performance, enabling users to examine revenue, costs, profitability, and sales trends.

The dataset contains 10,000 synthetic sales records covering a one-year period. The dashboard presents key performance indicators and visualizations covering monthly performance, geographical regions, product categories, sales representatives, sales channels, and customer segments.

The dashboard reports total revenue of KES 632,286,142.50, total profit of KES 149,687,444.40, total costs of KES 482,598,698.10, and a profit margin of 23.67%.

The project demonstrates the use of spreadsheet-based data analysis, performance measurement, interactive filtering, and data visualization.

2. Introduction

Sales data can help organizations understand how their products perform, where revenue is generated, and how costs affect profitability. However, large datasets can be difficult to interpret when presented only as rows and columns.

This project addresses that challenge by presenting sales information through an interactive Excel dashboard. It combines summary indicators, charts, and filters to make the data easier to explore and interpret.

The project is intended for educational and portfolio demonstration purposes. The dataset is synthetic and does not represent the verified operations of a real company.

3. Problem Statement

Raw sales records do not always provide a clear and immediate picture of business performance. Users may need to perform repeated calculations and comparisons to understand revenue trends, profitability, and differences across business categories.

The project therefore aims to organize sales records into a dashboard that makes key performance indicators and comparisons easier to access.

4. Project Objectives

The main objective was to develop an interactive sales dashboard using Microsoft Excel.

The specific objectives were to:

1. Summarize total revenue, total costs, total profit, and profit margin.
2. Examine monthly revenue and profit trends.
3. Compare performance across geographical regions and product categories.
4. Analyze sales representative performance.
5. Explore sales channels and customer segments.
6. Provide interactive filters for investigating different subsets of the data.
7. Present the results in a clear and accessible format.

5. Dataset Description

The dataset contains 10,000 synthetic retail sales records covering 1 January 2025 to 31 December 2025.

The data includes order dates, counties, regions, customer segments, products, product categories, sales representatives, sales channels, payment methods, quantities, unit prices, unit costs, revenue, total costs, profit, and profit margin.

The dataset was designed to support practice in data analysis and dashboard development. Its values are illustrative rather than verified business transactions.

Data preparation

The dataset was organized into a structured table so that the records could be analyzed consistently in Excel.

Before publication, the workbook and dataset should be checked for missing values, duplicate records, inconsistent categories, incorrect data types, and calculation errors. Any cleaning or transformation actually performed should be documented here.

6. Tools and Methodology

Microsoft Excel was used to organize the dataset, calculate performance indicators, and develop the dashboard.

The project workflow consisted of the following stages:

1. **Data organization:** Structuring the sales records into a table.
2. **Data validation:** Checking data types, values, categories, and calculations.
3. **Performance measurement:** Summarizing revenue, costs, profit, and profit margin.
4. **Data analysis:** Comparing performance across dates and business dimensions.
5. **Visualization:** Presenting the results through charts and KPI cards.
6. **Dashboard development:** Combining the visualizations into an interactive interface.
7. **Review:** Checking the accuracy, readability, and functionality of the final dashboard.

Only describe specific cleaning operations, formulas, or analytical techniques as completed if they were actually performed in the workbook.

7. Dashboard Design and Features

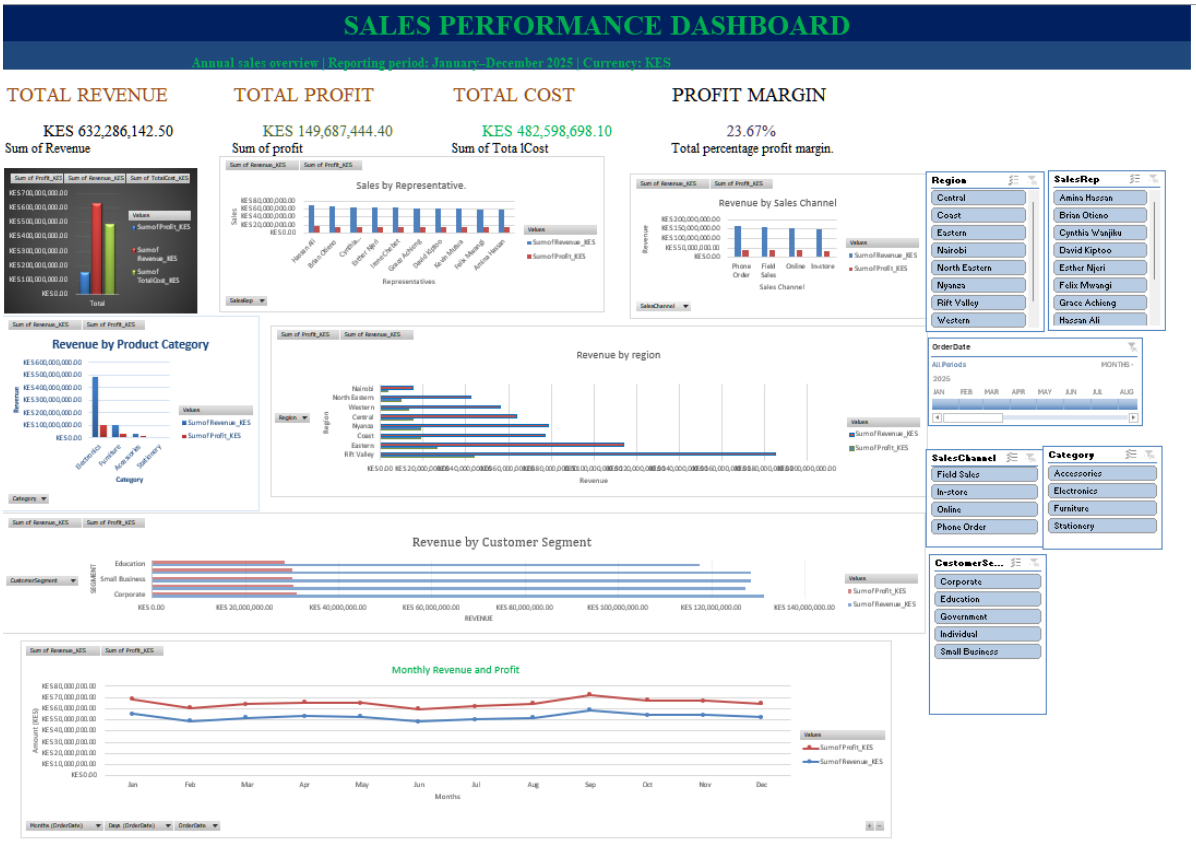
The dashboard contains four main performance indicators:

- Total Revenue
- Total Profit
- Total Cost
- Profit Margin

It also contains visualizations for regional performance, sales channels, product categories, monthly revenue and profit, sales representatives, and customer segments.

Interactive slicers allow users to filter by region, sales representative, sales channel, category, and customer segment. A date timeline supports exploration over the reporting period.

These features allow users to investigate selected parts of the dataset without manually searching through thousands of rows.



Sales Performance Dashboard developed using Microsoft Excel.

8. Key Results

The dashboard displays the following aggregate results:

Key Performance Indicator	Reported value
Total revenue	KES 632,286,142.50
Total costs	KES 482,598,698.10
Total profit	KES 149,687,444.40
Profit margin	23.67%

The displayed profit margin represents profit relative to revenue. The aggregate values should be verified against the final workbook before publication.

Interpretation

The dashboard provides a consolidated view of revenue, costs, and profitability. The relationship between these indicators allows users to understand the overall financial picture of the synthetic dataset.

The charts support further comparison across months, regions, categories, representatives, channels, and customer segments.

Specific claims about the best-performing region, most profitable category, strongest sales representative, or highest-performing month should be added only after reviewing the actual chart results.

9. Business Insights and Recommendations

The dashboard can support several types of business investigation:

1. **Regional performance:** Compare revenue and profit across regions to identify differences that require further investigation.
2. **Product categories:** Examine which categories contribute most to revenue and which deliver stronger profitability.
3. **Monthly trends:** Review monthly changes to identify possible growth periods, declines, or seasonal patterns.
4. **Sales representatives:** Compare performance while considering differences in sales volume, customer allocation, and profitability.
5. **Sales channels:** Evaluate how the different channels contribute to overall performance.
6. **Customer segments:** Compare segment contributions to identify areas for further customer analysis.

These are analytical recommendations, not confirmed findings about a real business. Specific actions should be based on verified results and, in a real organization, additional operational evidence.

10. Limitations

The dataset is synthetic, so its patterns may not reflect actual retail markets or customer behavior.

The dashboard summarizes the fields available in the dataset and cannot explain causes that were not recorded. For example, a difference in sales performance does not, by itself, establish why the difference occurred.

The results also depend on the accuracy of the underlying data, formulas, and dashboard calculations. The project should therefore be validated before being used for any real business decision.

11. Conclusion

This project demonstrates how Microsoft Excel can be used to organize sales data and communicate performance indicators through an interactive dashboard.

The resulting dashboard brings together revenue, costs, profitability, and comparisons across several business dimensions. It provides a practical demonstration of spreadsheet-based data analysis, data visualization, and business-oriented reporting.

The project also provides a foundation for future development using tools such as SQL, Power BI, or Python.

12. Project Deliverables

The project package is intended to include:

- Excel dashboard workbook
- CSV sales dataset
- README documentation
- Dashboard preview image
- Project report in PDF format
- Data dictionary

13. Data Disclaimer

All sales records and associated financial figures used in this project are synthetic and intended for educational and portfolio demonstration purposes. They should not be interpreted as actual transactions or financial results from a real organization.